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TEXTURE CLASSIFICATION USING TEXTURE SPECTRUM

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Abstract –Pursuing our previous study where the Texture Spectrum method has been proposed for texture analysis, the purpose of this paper is to demonstrate the usefulness of the Texture Spectrum for texture classification. Promising results are obtained when applying the Texture Spectrum to classify four of Brodatz's natural images.

Texture analysis Texture Unit Texture Spectrum Classification
Pattern recognition Brodatz

1. INTRODUCTION

Texture is the term used to characterize the surface of a given phenomenon in an image and it is undoubtedly one of the principal features used in image processing and pattern recognition. Texture classification plays an important role in the interpretation and analysis of terrain images, as well as radiographic and microscopic cell images.

Methods of texture classification can be broadly divided into two major categories. The first is texture segmentation, based on features with a high degree of spatial localization, e.g. with some form of edge detection. The basic assumption in this approach is that the intensity variation is more or less constant within the texture region and takes a greater value at its boundary. There are many known edge detection algorithms based on the gradient operator or zero-crossings of smoothed Laplacian operator such Robert, Sobel and Prewitt operators,[1] Marr and Hildreth's Laplacian-Gaussian operator[2] and Haralick's directional derivatives method.[3] The major difficulty for this kind of approach is in distinguishing between the texture boundaries which delimit different textures and the micro-edges located within the same texture. It is probably due to the lack of a formal definition of texture and of texture boundaries. In fact, the mechanism of the human visual system for texture discrimination is not yet very well understood.[4,5]

The second major approach to texture classification is based on a discrimination function using several texture characteristics. The linear maximum likelihood discrimination function is widely employed because of its easy implementation. In this method, the first and most important task is to extract texture features which most completely embody information of texture in the original image. Over the past few years, many methods have been proposed to extract texture features either directly from the image statistics, e.g. co-occurrence matrix, or from the spatial frequency domain.[6–9] The features derived from the co-occurrence matrix[8–10] and the features extracted from the most prominent components of the Fourier transformation[11,12] or from the image transformed by directional or isotropic filtering[13,14] have been used in practice for texture classification. Recognition scores in the range of 80–90% are frequently quoted in the literature.[10] Usually, the computing time required for extraction of texture features is excessive,[7] and most discriminant features are highly specific to each particular classification problem.[10]

Recently, He and Wang have introduced a new model of texture analysis based on the proposed concept of a Texture Unit, where a texture image can be characterized by its Texture Spectrum.[15] The aim of the present work is to further evaluate the performance of the Texture Spectrum in the discrimination of texture and demonstrate its usefulness and potential for texture classification.

2. TEXTURE UNIT AND TEXTURE SPECTRUM

This section gives a brief review of the Texture Unit and the Texture Spectrum, which have been

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introduced and described in detail in the previous works.[15]

The basic concept is that a texture image can be decomposed into a set of essential small units, called Texture Units. A Texture Unit is represented by eight elements, each of which has one of three possible values (0, 1, 2) obtained from a neighbourhood of 3×3 pixels. Figure 1 shows an example of transforming a neighbourhood to a Texture Unit. In total, we have $3^8 = 6561$ standard Texture Units. Such a defined Texture Unit is considered as the smallest complete unit which best characterizes the texture aspect in all the eight directions from the central pixel considered. The processing of intensity values into only three values (0, 1, 2) reduces the calculation time for the pursuit statistics, and also eliminates the undesirable influence of both the regional intensity background along with noise in the original image.

As the Texture Unit represents the local texture aspect, the statistics of Texture Units in an image should reveal its texture information. The occurrence distribution of Texture Units is called the Texture Spectrum, with the abscissa indicating the type of Texture Unit and the ordinate representing its occurrence frequency. Figure 2 gives an example of Texture Spectrum for the four images of Fig. 3.

The previous work has demonstrated that different texture images have correspondingly different spectra; in the other words, the texture aspect of an image can be characterized by its spectrum.

3. TEXTURE CLASSIFICATION

As previously mentioned, the most important problem in texture classification is to extract powerful features which best characterize the texture aspect of image. In order to evaluate the discriminating performance of Texture Spectrum, a classification has been performed over four texture images extracted from Brodatz's album.[16] These natural images have been digitized with the M.A.M.A. machine.[17] Each image is represented by 256×256 pixels with 64 normalized grey levels (Fig. 3). They are respectively the image of (A) beach sand, (B) water, (C) pressed cork and (D) fur hide of unborn calf.

To demonstrate the pure and original property of the Texture Spectrum, we use the simplest discrimination method and carry out as few as possible

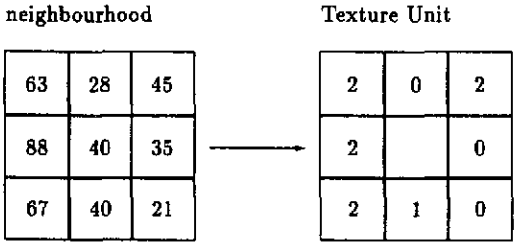


Fig. 1. Example of transforming a neighbourhood to a Texture Unit.

mathematical operations over the Texture Spectrum. The evaluation was performed using a supervised classification over the four texture images of Fig. 3. The four textures are classified to four classes using the minimum distance decision rule. The procedure of the classification is described as follows:

(1) Select randomly a sample subimage of 30×30 pixels from each texture image (one sample per texture);

(2) Calculate Texture Spectrum for each sample subimage, by moving the 3×3 matrix across the sample with overlap;

(3) Scan the four textures of Fig. 3 by a window of 30×30 pixels with a step of two pixels in the row and column directions, and calculate the Texture Spectrum for each window;

(4) Calculate the absolute difference between the Texture Spectrum of each window and one of each sample:

$$D(i) = \sum_{j=1}^{6561} |W(j) - S(i, j)|, \quad i = 1, 2, 3, 4$$

where:

$D(i)$ denotes the absolute difference between the Texture Spectrum of a window and the Texture Spectrum of a sample subimage i; $W(j)$ represents the occurrence value of Texture Unit j in the Texture Spectrum of the window considered; $S(i, j)$ represents the occurrence value of Texture Unit j in the Texture Spectrum of the sample subimage i.

(5) The central pixel of the window considered will be assigned to class K such that $D(K)$ is minimum among all the $D(i)$, for $i = 1, 2, 3, 4$.

4. RESULTS AND EVALUATION

Using the above described method, $58081 = ((512 - 30)/2)^2$ pixels of Fig. 3 have been processed and assigned to one of the four classes. The result, illustrated in Fig. 4 where the four class numbers are represented by four different grey levels, shows 98.5% correct classification for texture image A, 96.4% for texture B, 98.9% for texture C and 95.9% for texture D, resulting in the averaged recognition rate of 97.5%.

It should be noted that the above correct classification rates are calculated over all the pixels including the region near the boundaries of the four textures. If we remove these pixels from the counter, the correct classification rate will be 100% for the textures (A), (B) and (C), and 98.4% for (D).

In addition to the above quantitative analysis of the classification result, we have calculated the probability of assigning a pixel to its class, which can be defined as:

$$P = \frac{1 - D(k)}{\sum_{j=1}^{4} D(j)}$$

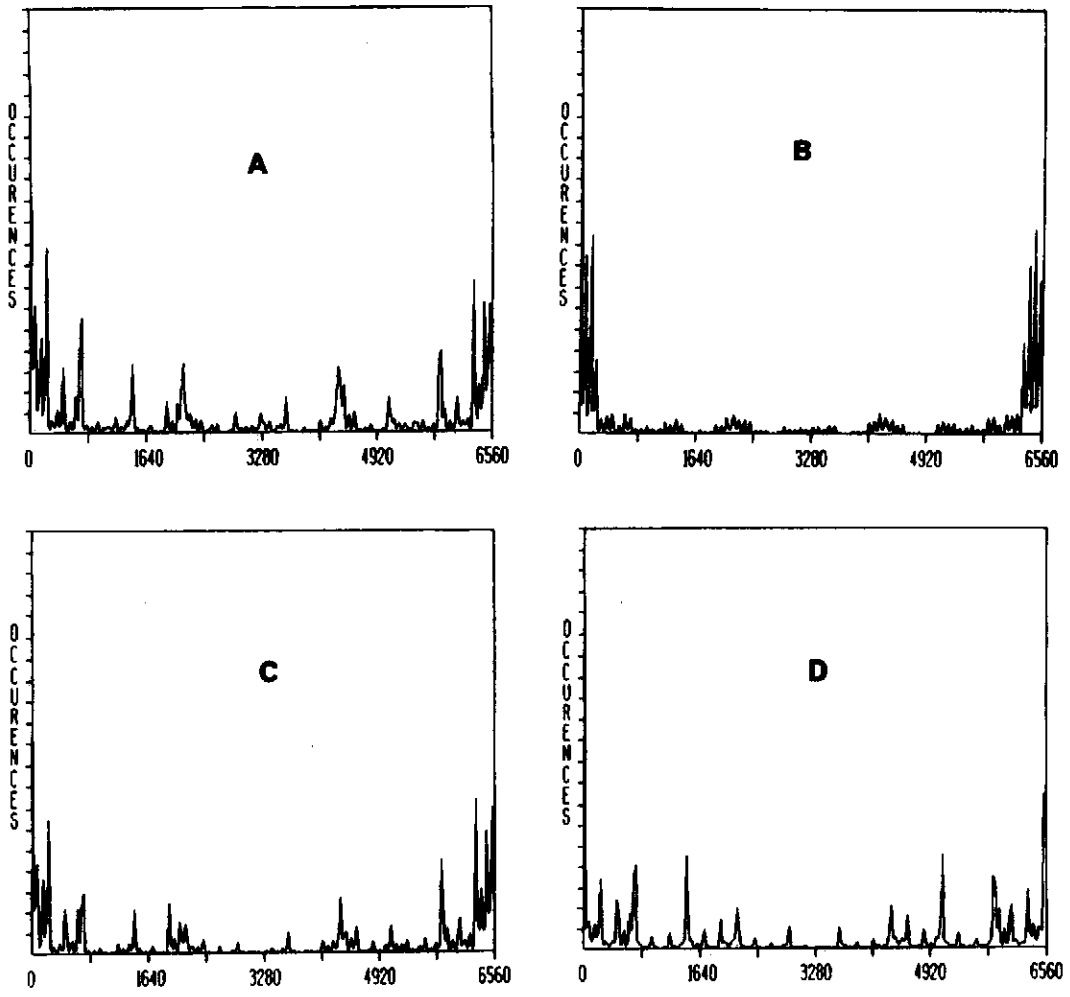


Fig. 2. Example of Texture Spectrum calculated from the images of Fig. 3.

where:

$$D(k) \in D(j) \text{ for } j = 1,2,3,4 \text{ and } k \in 1,2,3,4$$

Figure 5 illustrates the calculated probability P for each element of Fig. 4, where the brightness is inversely proportional to the probability (P). We can note that:

(1) Between the texture (A) and (B) and between (B) and (D) we find an obvious white boundary indicating a large difference between them. At these boundaries, the scanning window crosses two kinds of textures, resulting in a mixed Texture Spectrum and so giving a lower value of the probability. In fact, the mixed Texture Spectrum is composed of the Texture Spectrum of each texture weighted by the area occupied in the window. The greater the proportion of one texture compared to the other, the closer will be the mixed Texture Spectrum to that of one image. Thus the Texture Spectrum could reveal, to a certain degree, information about the proportion of different textures within a window.

(2) The up-right part of Fig. 5 is a relatively homogeneous black region, illustrating that the texture (B) has been classified into class 2 with a relatively constant higher probability. It should be noted that, the original texture (B) has a large background intensity variation.

(3) Texture D has been classified into the class 4 with a large deviation of probability. In fact, texture D is not homogeneous if we consider its directional orientation.

(4) The four sample subimages have been arbitrarily selected in Fig. 3. The two black discs in the left part of Fig. 5 indicate the higher value of the probability around the position of the two samples of the texture (A) and (C).

5. CONCLUSION AND DISCUSSION

The Texture Spectrum has been evaluated from the point of view of discriminating performance. Four of Brodatz's natural images have been classified using the Texture Spectrum. The discrimination method was simple, and the number of mathematical operations applied to the Texture Spectrum was small, while promising results have been obtained with an average correct classification rate of 97.5% including

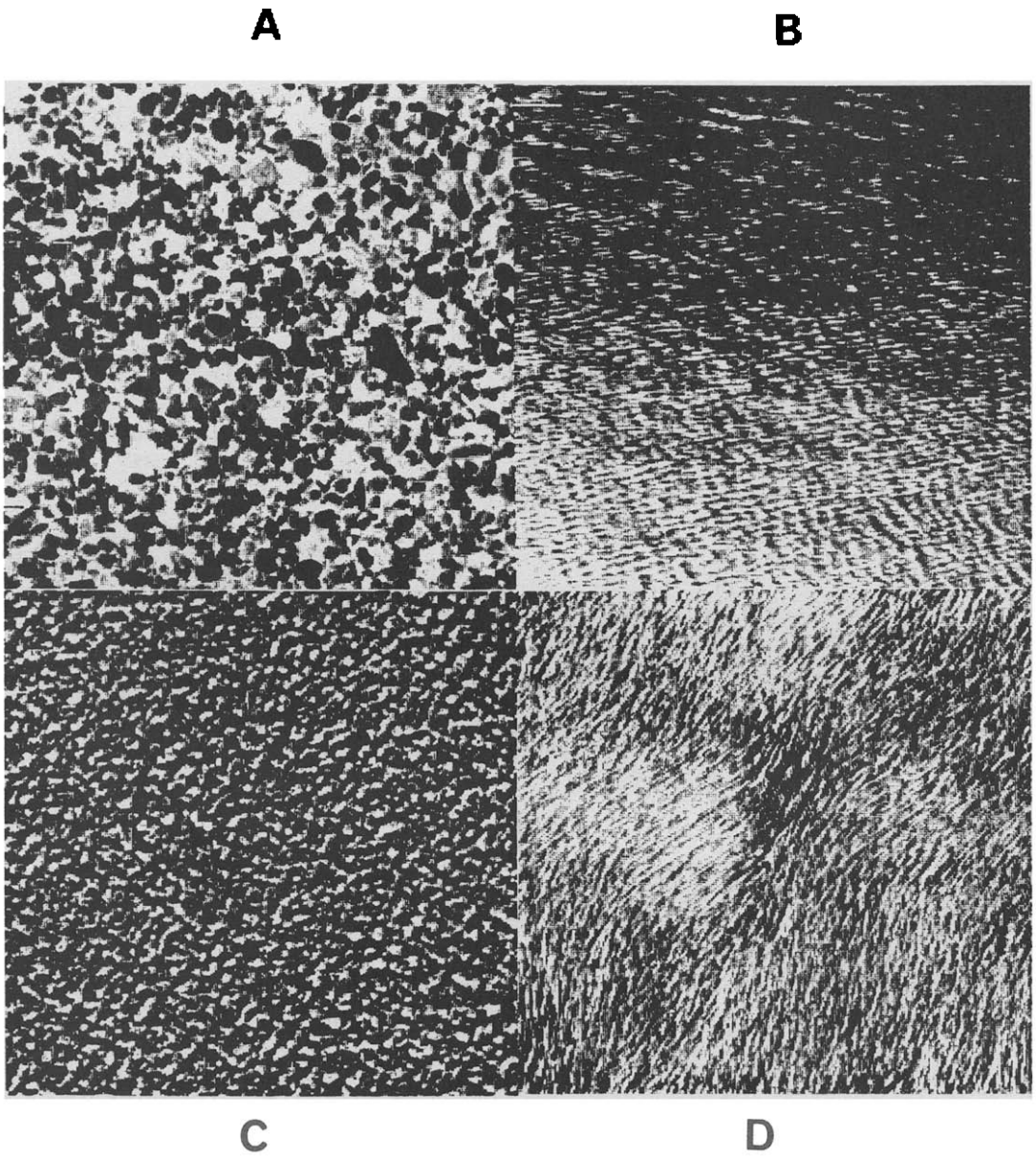


Fig. 3. Four of Brodatz' texture images: (A) beach sand, (B) water, (C) pressed cork and (D) fur hide of unborn calf.

the influence of the boundaries of different textures.

Further evaluation of section 4 shows that the Texture Spectrum is sensitive to the directional aspect of texture (Fig. 5D). In addition, the undesirable influence of the regional intensity background is eliminated from the Texture Spectrum (Fig. 5B). So, the study of the proper similarity of texture can be conducted.

The Texture Spectrum method has been recently proposed for texture analysis. The work presented in this paper is just a preliminary study to evaluate the performance of the Texture Spectrum in the discrimination of textures. Although the features derived from the Texture Spectrum have not yet been proposed, such an evaluation is possible and would be of use in the recognition of the Texture Spectrum. By only using the difference between the Texture Spectra, one is already able to distinguish between several different textures, lending support to the use of the Texture Spectrum method for texture analysis. Of course, it is far from constituting a complete, independent analysis method. Some further study is required, including the extraction of texture features from the Texture Spectrum in order to more efficiently carry out the texture classification.

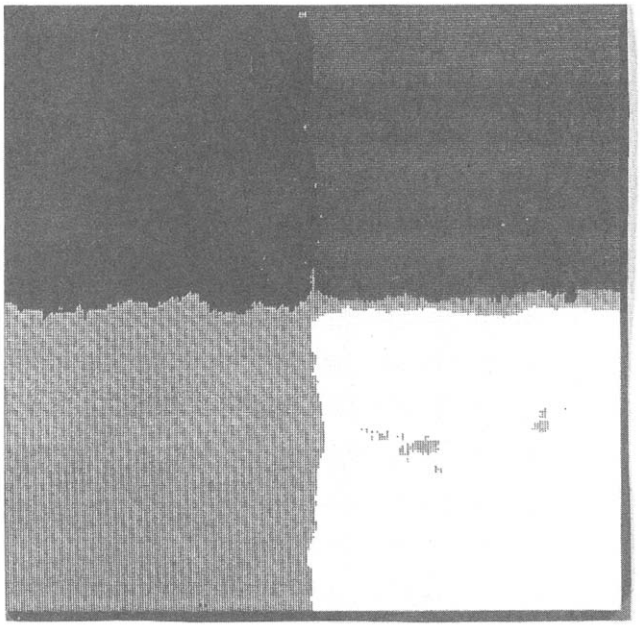


Fig. 4. Supervised classification of Fig. 3. The four classes are represented by four grey levels.

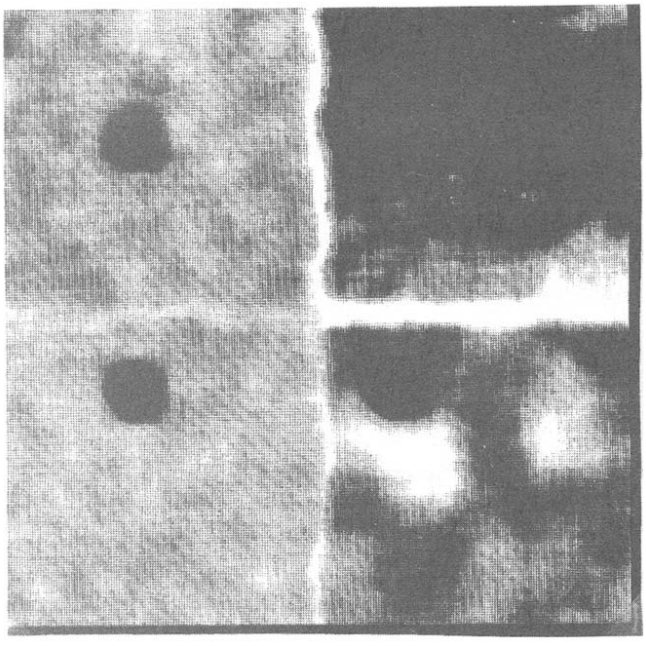


Fig. 5. Assigning Probability of classification for each element of Fig. 4. The brightness is inversely proportional to the probability P.

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